SDIE v3.3 Enterprise Implementation Package

Project: Inizio Structural Component Intelligence Engine

Strategic Objective

Replace the current Geometry-First Slab Detection Architecture with a Structural Component Intelligence Architecture capable of learning from tagged structural drawings and estimator outputs.

Target:

DXF → Structural Understanding → Component Classification → Semantic Building Model → Quantity Estimation

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PART 1 – CURRENT SYSTEM GAP ANALYSIS

Current SDIE Pipeline:

DXF ↓ Beam Grid ↓ Grid Cells ↓ THK Merge ↓ Slab Candidates ↓ Quantity

Observed Limitation:

The engine does not understand:

- Beam
- Column
- Shear Wall
- Lift Core
- Stair Core
- Shaft
- Structural Wall

Therefore non-slab regions are incorrectly classified as slabs.

Root Cause:

Current engine solves geometry before semantics.

Required Change:

Semantics must precede geometry-based quantity estimation.

PART 2 – INIZIO STRUCTURAL COMPONENT ATLAS

Create a reusable Component Atlas from tagged drawings.

Classes:

1. Slab
2. Beam
3. Column
4. Shear Wall
5. Structural Wall
6. Lift Core
7. Stair Core
8. Shaft
9. Opening
10. Unknown

Atlas Sample Schema

{ "sample_id": "", "project_id": "INIZIO", "component_type": "", "geometry_features": {}, "annotation_features": {}, "graph_features": {}, "source_drawing": "", "confidence": 1.0 }

Atlas becomes the foundation dataset for all future learning.

PART 3 – AUTO LABELLING ENGINE

Inputs:

- Beam tagged DXF
- Slab tagged DXF
- Column tagged DXF
- Shearwall tagged DXF

Process:

Tagged Drawing ↓ Extract Tagged Entity ↓ Convert to Geometry ↓ Generate Features ↓ Create Atlas Sample

Output:

Atlas Database

No manual labelling after initial project setup.

PART 4 – COMPONENT CLASSIFIER

Classifier Input:

- Geometry
- Topology
- Layer Metadata
- Annotation Text
- Graph Relationships

Classifier Output:

{ "component":"Beam", "confidence":0.96 }

Target Accuracy:

95%+

PART 5 – FEATURE ENGINEERING

Beam Features

- Length
- Width
- Aspect Ratio
- Connectivity
- Repetition

Column Features

- Rectangular Geometry
- Grid Pattern
- Beam Support Count

Wall Features

- Thickness
- Continuity
- Core Proximity

Slab Features

- Beam Enclosure
- THK Annotation
- Area
- Exclusion Relationship

PART 6 – STRUCTURAL GRAPH ENGINE

Node Types

- Slab
- Beam
- Column
- Wall
- Core

Relationships

- Supports
- Frames
- Touches
- Contains
- Adjacent

Example

Column ↓ Beam ↓ Slab

Graph stored in NetworkX.

Future ready for GNN.

PART 7 – DEEPSEEK REASONING LAYER

DeepSeek Responsibilities

Allowed:

- Component Classification
- Ambiguity Resolution
- Missing Annotation Reasoning

Forbidden:

- Area Calculation
- Volume Calculation
- Quantity Estimation

Output Schema

{ "classification":"", "confidence":0.0, "evidence":[] }

PART 8 – SEMANTIC BUILDING MODEL

Building ├ Floors ├ Zones ├ Components ├ Relationships └ Quantities

JSON Representation

{ "building":{ "floors":[], "components":[], "relationships":[] } }

PART 9 – SLAB INTELLIGENCE ENGINE

New Logic

Identify:

Beam Column Wall Core

First.

Then:

Remaining framed region

Potential Slab

Exclusions:

- Staircases
- Lift Cores
- Shafts
- Structural Walls
- Openings

PART 10 – BENCHMARK FRAMEWORK

Ground Truth Source

Estimator Workbook

Available Fields

- Slab ID
- Area
- Thickness
- Concrete

• Shuttering

Benchmark Metrics

Area Accuracy Thickness Accuracy Concrete Accuracy Shuttering Accuracy

Target:

95%+

PART 11 – CONFIDENCE FRAMEWORK

Confidence Score

0.35 × Geometry

+ 0.25 × Topology

+ 0.20 × Graph

+ 0.20 × DeepSeek

Final Confidence:

0–100%

PART 12 – FASTAPI ARCHITECTURE

src/

ingestion/ atlas/ classification/ graph/ reasoning/ quantity/ validation/ api/ database/

PART 13 – DATABASE DESIGN

Projects Drawings Floors Zones Components Relationships AtlasSamples Benchmarks Quantities

Database:

PostgreSQL

PART 14 – REACT APPLICATION

Screens

1. Project Dashboard
2. Drawing Viewer
3. Component Explorer
4. Atlas Manager
5. Validation Center
6. Benchmark Reports

PART 15 – CURSOR IMPLEMENTATION ROADMAP

Epic 1 Atlas Builder

Acceptance: Tagged drawings converted to atlas samples

Epic 2 Component Classifier

Acceptance: 95% classification accuracy

Epic 3 Structural Graph

Acceptance: Relationships generated

Epic 4 Semantic Building Model

Acceptance: Building model export working

Epic 5 Slab Intelligence

Acceptance: Non-slab regions excluded

Epic 6 Quantity Engine

Acceptance: 95% concrete accuracy

Epic 7 Validation UI

Acceptance: Human review workflow operational

FINAL DIRECTIVE

Do not improve slab detection directly.

Build Structural Component Intelligence first.

All slab estimation must occur only after component classification and semantic building model generation.